A COEFFICIENT PROBLEM IN H^p WITH PRESCRIBED VALUE AT THE ORIGIN: ZEROS AND THE TERMINAL INTERVAL

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ABSTRACT. For $0 < p < 1$, $k \geq 1$ and $0 \leq t \leq 1$, let $\Phi_k(p,t)$ denote the supremum of $\operatorname{Re} f^{(k)}(0)/k!$ over functions f in the unit ball of the Hardy space H^p with $f(0) = t$. Brevig, Grepstad and Instanes asked whether the extremals have at most one zero in the unit disc, and whether $\Phi_k(p,t) = \Phi_1(p,t)$ when t is close to 1. We answer both questions affirmatively for every k. We also show that, for fixed p, the maximal interval near $t = 1$ on which $\Phi_k = \Phi_1$ has length of order $k^{-2(1-p)/(2-p)}$ as $k \to \infty$.

1. INTRODUCTION

Let H^p denote the classical Hardy space of analytic functions in the unit disc $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, which for $0 < p < \infty$ is equipped with

$$\|f\|_{H^p}^p = \lim_{r \to 1^-} \int_0^{2\pi} |f(re^{i\theta})|^p \, \frac{\mathrm{d}\theta}{2\pi}.$$

Suppose that $k \geq 1$ is an integer and that $0 \leq t \leq 1$. Brevig, Grepstad and Instanes [2] considered the extremal problem

$$\Phi_k(p,t) = \sup\Big\{ \operatorname{Re} \frac{f^{(k)}(0)}{k!} : \|f\|_{H^p} \leq 1 \text{ and } f(0) = t \Big\}. \tag{1}$$

By a standard normal families argument, there are functions attaining the supremum in (1), which we call extremals, and every extremal satisfies $\|f\|_{H^p} = 1$ [2, Lemma 3]. When $t = 0$, the problem (1) reduces, through $f = zF$, to the coefficient problem $C(k-1,p)$ studied by Brevig and Saksman [3].

In the range $1 < p \leq \infty$, the extremal has either 0 or k zeros in $\mathbb{D}$, while for $p = 1$ any number between 0 and k can occur [2, Theorem 15]. For $0 < p < 1$ and $k = 1$, the problem was studied by Kabaila [7, 8] and Connelly [4, Section 4], and solved completely in [2, Theorem 12]. There is $0 < t_p < 1/2$ such that the extremal has one zero in $\mathbb{D}$ when $t < t_p$ and none when $t > t_p$, and Kabaila showed that t_p is the only value of t for which there

Date: October 10, 2026.

2020 *Mathematics Subject Classification.* Primary 30H10.

Key words and phrases. Hardy spaces, linear extremal problems, coefficient estimates, zeros of extremal functions.

are two extremals, one with a zero and one without. Moreover, if $t_p \le t \le 1$ and $t = (1+\beta^2)^{-1/p}$ with $0 \le \beta \le 1$, then

$$\Phi_1(p,t) = \frac{2\beta}{p}\,\frac{1}{(1+\beta^2)^{1/p}},$$

and the extremal is $(1+\beta z)^{2/p}(1+\beta^2)^{-1/p}$. For $k \ge 2$, Brevig, Grepstad and Instanes asked the following two questions [2, Section 6].

Question 1. *Suppose that $0 < p < 1$. Is it true that the extremal for $\Phi_k(p,t)$ has at most one zero in $\mathbb{D}$?*

Question 2. *Fix $k \ge 2$ and $0 < p < 1$. Is there some t_0 such that $\Phi_k(p,t) = \Phi_1(p,t)$ holds for every $t_0 \le t \le 1$?*

They verified numerically that the answer to Question 1 is affirmative for $k = 2$, and gave strong evidence for an affirmative answer to Question 2 for $k = 2$. The main purpose of the present paper is to answer both questions affirmatively for every k. Our first result is the following.

Theorem 1. *Suppose that $0 < p < 1$, $k \ge 1$ and $0 \le t \le 1$. Then every extremal for $\Phi_k(p,t)$ has at most one zero in $\mathbb{D}$, counting multiplicity.*

Our second result answers Question 2, and it also identifies the length of the interval on which $\Phi_k = \Phi_1$.

Theorem 2. *Suppose that $0 < p < 1$, and put $\alpha = 2(1-p)/(2-p)$. There are constants $c, C > 0$, depending only on p, such that the following hold.*

(a) *If $k \ge 1$ and $1 - ck^{-\alpha} \le t \le 1$, write $t = (1+\beta^2)^{-1/p}$. Then*

$$\Phi_k(p,t) = \Phi_1(p,t) = \frac{2\beta}{p}\,\frac{1}{(1+\beta^2)^{1/p}},$$

and the unique extremal for $\Phi_k(p,t)$ is $(1+\beta z^k)^{2/p}(1+\beta^2)^{-1/p}$.

(b) *If k is sufficiently large and $t = 1 - Ck^{-\alpha}$, then $\Phi_k(p,t) > \Phi_1(p,t)$.*

In particular, the answer to Question 2 is affirmative for every k. If we define the terminal threshold

$$T_k(p) = \inf\{a : \Phi_k(p,t) = \Phi_1(p,t) \text{ for } a \le t \le 1\},$$

then Theorem 2 states that $1 - T_k(p) \asymp_p k^{-\alpha}$ as $k \to \infty$. Note that Theorem 2 determines the order of $1 - T_k(p)$, but not the leading constant (see Section 6.2).

The bound in Theorem 1 cannot be improved, since the extremals have exactly one zero when t is small.

Corollary 1. *Suppose that $0 < p < 1$ and $k \ge 1$. There is $t_0 = t_0(k,p) > 0$ such that every extremal for $\Phi_k(p,t)$ has exactly one zero in $\mathbb{D}$ when $0 \le t < t_0$.*

For $t = 0$ and $k \geq 2$, Theorem 1 states that every extremal zF has exactly one zero, so the extremals F for $C(k-1, p)$ do not vanish in $\mathbb{D}$. This is Conjecture 1 of Brevig and Saksman [3, Section 5], which was proved for all bounded linear functionals in [10]. The argument of the present paper gives an independent proof for the coefficient functionals.

Let us briefly describe our approach to Theorem 1. As in [1, 3], we factor an extremal as $f = gh^{2/p-1}$, where g and h are in the unit sphere of H^2, the function h is outer, and g/h is inner. For a fixed h, the prescribed value $f(0) = t$ fixes $g(0)$, and the Cauchy–Schwarz inequality determines the best remaining first factor. This turns the problem into a non-linear extremal problem for h alone. Brevig and Saksman [3, Lemma 3] perturb h to $(1+\varepsilon\varphi)h$ and use the first derivative in ε, adapting a variational argument of F. Riesz [9]. If f has two zeros, we use them to construct a direction φ with $\varphi(0) = 0$. Normalizing the perturbed h changes $h(0)$, and the reduced problem then adjusts $g(0)$ so that $f(0) = t$ is kept. In this direction the first derivative vanishes, and the second derivative is a positive multiple of

$$(q-1)\int_0^{2\pi} \left(\operatorname{Re}\varphi(e^{i\theta})\right)^2 |h(e^{i\theta})|^2 \,\frac{\mathrm{d}\theta}{2\pi}, \qquad q = \frac{2}{p} - 1.$$

This is positive when $p < 1$, so f is not an extremal, while for $p = 1$ it vanishes.

The scale in Theorem 2 comes from a competition between two kinds of functions. Near $t = 1$, the constant term uses almost all of the norm, and only a budget of order $\delta = 1 - t$ remains for the other coefficients. Spent directly on the coefficient of z^k, through the binomial $(1 + \beta z^k)^{2/p}$, it yields a coefficient of order $\sqrt{\delta}$. Spent instead on a function concentrated near a point of the circle, such as $(1-\rho z)^{-2/p}$ with ρ close to 1, it yields a coefficient of order $\delta^{1/p}k^{1/p-1}$. The two balance when $\delta \approx k^{-\alpha}$, and part (b) shows that concentration wins beyond this scale. For part (a), the idea is to split f into $m = \lceil 2/p \rceil$ factors in H^r with $2 \leq r < 3$, and to compare each factor with the binomial model maximizer. Uniform convexity of $|\cdot|^r$ makes every deviation from it cost a definite amount, and concentration can only help through products of at least two deviations, which are too small below the scale $k^{-\alpha}$.

Organization. The present paper is organized into five additional sections. In Section 2 we reduce (1) to an extremal problem for h and obtain the structure of the extremals. Section 3 contains the proof of Theorem 1, and Section 4 contains the proof of Theorem 2. In Section 5 we prove Corollary 1, and Section 6 contains some concluding remarks.

2. Optimizing the first factor

Throughout the paper we assume that $0 < p < 1$ and set $q = 2/p - 1$, so that $q > 1$. We identify H^2 with a subspace of $L^2(\mathbb{T})$, where $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, and write $\langle u, v\rangle = \int_0^{2\pi} u(e^{i\theta})\overline{v(e^{i\theta})}\,\mathrm{d}\theta/2\pi$. For $u(z) = \sum_{n\geq 0} u_n z^n$

we write $S_m u(z) = \sum_{n=0}^{m} u_n z^n$, and for a polynomial u of degree at most k we write $u^*(z) = z^k \overline{u(1/\overline{z})}$, so that $|u^*| = |u|$ on $\mathbb{T}$. For $\varphi \in L^\infty(\mathbb{T})$ and $h \in H^2$ we write $\langle \varphi, |h|^2 \rangle = \int_0^{2\pi} \varphi |h|^2 \, \mathrm{d}\theta/2\pi$, which equals $\langle \varphi h, h \rangle$. For a non-vanishing analytic function h in $\mathbb{D}$, h^q denotes a fixed branch, and powers of perturbations $(1+\varepsilon\varphi)h$ with $\|\varepsilon\varphi\|_{H^\infty} < 1$ are the branches $(1+\varepsilon\varphi)^q h^q$, with the principal branch of the first factor. Hölder's inequality gives

$$\|gh^q\|_{H^p} \le \|g\|_{H^2} \|h\|_{H^2}^q \tag{2}$$

for g, $h \in H^2$ with h non-vanishing, since $1/p = 1/2 + q/2$.

In preparation for the proof of Theorem 1, we first optimize over the first factor when the second factor is fixed. Fix $k \ge 1$ and $0 \le t < 1$. Suppose that h is a non-vanishing function in the unit sphere of H^2 with $|h(0)|^q > t$, and let $(c_n)_{n\ge 0}$ be the coefficient sequence of h^q. If g has coefficient sequence $(b_n)_{n\ge 0}$, then

$$\frac{(gh^q)^{(k)}(0)}{k!} = \sum_{j=0}^{k} b_j c_{k-j}, \qquad (gh^q)(0) = b_0 c_0. \tag{3}$$

The condition $(gh^q)(0) = t$ fixes $b_0 = t/c_0$, and by the Cauchy–Schwarz inequality the largest value of the real part of (3) over g in the unit ball of H^2 with $b_0 = t/c_0$ is

$$\Lambda(h) = \operatorname{Re} \frac{t c_k}{c_0} + \Big(1 - \frac{t^2}{|c_0|^2}\Big)^{1/2} \Big(\sum_{j=0}^{k-1} |c_j|^2\Big)^{1/2}. \tag{4}$$

The first term comes from the prescribed constant term of g, and the second from the best choice of its remaining coefficients. The maximum is attained by a polynomial g of degree at most k, and since $\|gh^q\|_{H^p} \le 1$ by (2), we have

$$\Lambda(h) \le \Phi_k(p,t). \tag{5}$$

Suppose now that f is an extremal for (1). Every nonzero function in H^p can be written as $f = IO$, where I is an inner function and O is an outer function. This allows us to factor $f = gh^q$, where $h = O^{p/2}$ and $g = Ih$, and where the powers of O are taken with respect to one branch of $\log O$ so that $h^{q+1} = O$. We note that $|g| = |h| = |f|^{p/2}$ on $\mathbb{T}$, which yields $\|g\|_{H^2} = \|h\|_{H^2} = 1$. After multiplying g and h by unimodular constants, we may assume that $h(0) > 0$ and $c_0 = h(0)^q > 0$. Let $(b_n)_{n\ge 0}$ be the coefficient sequence of g, and put $s = b_0 = t/h(0)^q$. Since $h(0) \le \|h\|_{H^2} = 1$, we have $s \ge t$. Note also that the zeros of f in $\mathbb{D}$ are the zeros of g.

Lemma 1. *Suppose that $f = gh^q$ is an extremal for (1) with $0 \le t < 1$, and that g is not constant. Then $0 \le s < 1$, $\Lambda(h) = \Phi_k(p,t)$, and g is a polynomial of degree k. Moreover, there is a constant $\mu > 0$ such that*

$$\mu g^*(z) = h(z)^q + O(z^k). \tag{6}$$

Proof. Since g is not constant, we have $s < \|g\|_{H^2} = 1$. In particular, $c_0 = t/s > t$ when $t > 0$, and $c_0 > 0 = t$ when $t = 0$, so (5) applies to h. By (3), the Cauchy–Schwarz inequality and (5), we find that

$$\begin{aligned}\Phi_k(p,t) &= \operatorname{Re}(sc_k) + \operatorname{Re}\sum_{j=1}^{k} b_j c_{k-j} \\ &\le \operatorname{Re}(sc_k) + (1-s^2)^{1/2}\Big(\sum_{j=0}^{k-1}|c_j|^2\Big)^{1/2} = \Lambda(h) \le \Phi_k(p,t),\end{aligned}$$

where we used that $\sum_{j\ge1}|b_j|^2 = 1 - s^2$. Hence there is equality throughout. Equality in the Cauchy–Schwarz inequality forces $b_j = 0$ for $j > k$ and $\mu b_j = \overline{c_{k-j}}$ for $1 \le j \le k$, where $\mu = (1-s^2)^{-1/2}(\sum_{j=0}^{k-1}|c_j|^2)^{1/2} > 0$. This is (6), and $\overline{b_k} = c_0/\mu \ne 0$. □

Lemma 1 states that an extremal is determined by its second factor h, through the reduced objective Λ, and this is what allows us to perturb h alone in the next section. Equation (6) is the analogue of [3, (9)]. The error term is $O(z^k)$ rather than $O(z^{k+1})$, because b_0 is fixed by the condition $f(0) = t$. Note that $\|g^*\|_{H^2} = 1$, that $|g^*| = |h|$ on $\mathbb{T}$, and that the coefficient of z^k in g^* is $\overline{b_0} = s$. For $p = 1$, the same optimization appears in the proof of [2, Theorem 15(b)], after Riesz [9], and the polynomial structure of g also follows from [2, Lemma 4], which is based on [6].

3. Two zeros give an improving direction

Suppose that $f = gh^q$ is as in Lemma 1, and that g has two zeros λ_1, λ_2 in $\mathbb{D}$, counting multiplicity; one or both may be 0. Since g is a polynomial of degree k, we can write $g(z) = (z-\lambda_1)(z-\lambda_2)P(z)$ with $\deg P = k-2$, and reflecting gives

$$g^*(z) = W(z)U(z), \qquad W(z) = (1-\overline{\lambda_1}z)(1-\overline{\lambda_2}z),$$

where $U(z) = z^{k-2}\overline{P(1/\overline{z})}$ is a polynomial of degree at most $k-2$. The function W has no zeros in $\overline{\mathbb{D}}$. Dividing g^* by W and multiplying by z therefore gives a direction

$$\varphi(z) = \frac{\omega z}{W(z)}, \qquad \varphi g^* = \omega z U,$$

in which $\varphi \in H^\infty$, $\varphi(0) = 0$, and φg^* is a polynomial of degree at most $k-1$. After choosing $|\omega| = 1$ with $\operatorname{Re}(\omega\langle zU, g^*\rangle) = 0$, we also have $\operatorname{Re}\langle\varphi, |h|^2\rangle = \operatorname{Re}\langle\varphi g^*, g^*\rangle = 0$, since $|g^*| = |h|$ on $\mathbb{T}$. The next lemma is the key new ingredient in the proof of Theorem 1. It shows that such a direction cannot exist.

Lemma 2. *Suppose that $f = gh^q$ is as in Lemma 1, and that $\varphi \in H^\infty$ satisfies $\varphi(0) = 0$ and $\operatorname{Re}\langle\varphi, |h|^2\rangle = 0$, and that $v = \varphi g^*$ is a polynomial of*

degree at most $k-1$. For real ε close to 0, put $h_\varepsilon = (1+\varepsilon\varphi)h/\|(1+\varepsilon\varphi)h\|_{H^2}$. Then

$$\Lambda(h_\varepsilon) = \Lambda(h) + q(q-1)\mu\,\varepsilon^2 \int_0^{2\pi} (\operatorname{Re}\varphi)^2 |h|^2 \,\frac{\mathrm{d}\theta}{2\pi} + O(\varepsilon^3). \tag{7}$$

Consequently, $\varphi = 0$.

Proof. We begin with the consequence. Write $n_\varepsilon = \|(1+\varepsilon\varphi)h\|_{H^2}^2$. Since $\operatorname{Re}\langle\varphi h, h\rangle = 0$ and $|g^*| = |h|$ on $\mathbb{T}$, we have $n_\varepsilon = 1+\varepsilon^2\|\varphi h\|_{H^2}^2 = 1+\varepsilon^2\|v\|_{H^2}^2$. The function h_ε is in the unit sphere of H^2 and does not vanish in $\mathbb{D}$, and since $\varphi(0) = 0$, we have $h_\varepsilon(0)^q = n_\varepsilon^{-q/2} c_0 > t$ for ε close to 0. By (5) and Lemma 1, it follows that $\Lambda(h_\varepsilon) \le \Phi_k(p,t) = \Lambda(h)$, and so (7) gives $\int_0^{2\pi} (\operatorname{Re}\varphi)^2 |h|^2 \,\mathrm{d}\theta/2\pi = 0$. Since h is outer, $|h| > 0$ almost everywhere on $\mathbb{T}$. Hence the bounded harmonic function $\operatorname{Re}\varphi$ vanishes, φ is an imaginary constant, and $\varphi(0) = 0$ gives $\varphi = 0$. It remains to prove (7).

Let $(c_n(\varepsilon))_{n\ge 0}$ be the coefficient sequence of $((1+\varepsilon\varphi)h)^q$, so that the coefficients of h_ε^q are $n_\varepsilon^{-q/2} c_n(\varepsilon)$. Since $\varphi(0) = 0$, we have $c_0(\varepsilon) = c_0$, and (4) with $t/c_0 = s$ gives

$$\Lambda(h_\varepsilon) = \operatorname{Re}\big(s c_k(\varepsilon)\big) + \Big((n_\varepsilon^{-q} - s^2) \sum_{j=0}^{k-1} |c_j(\varepsilon)|^2\Big)^{1/2}. \tag{8}$$

Put $a = S_{k-1} g^*$, so that $g^* = a + sz^k$ and $R = \|a\|_{H^2} = \sqrt{1-s^2} > 0$, and put $B_\varepsilon = (1+\varepsilon\varphi)^q g^*$ and $X_\varepsilon = S_{k-1} B_\varepsilon$. By (6), the function $h^q - \mu g^*$ vanishes to order k at 0, and so does $(1+\varepsilon\varphi)^q(h^q - \mu g^*)$, with the same coefficient $c_k - \mu s$ of z^k, since $\varphi(0) = 0$. Hence

$$\sum_{j=0}^{k-1} |c_j(\varepsilon)|^2 = \mu^2 \|X_\varepsilon\|_{H^2}^2, \qquad c_k(\varepsilon) = \mu[z^k] B_\varepsilon + c_k - \mu s.$$

The binomial series of $(1+\varepsilon\varphi)^q$ converges in H^∞, so $X_\varepsilon = a + q\varepsilon v + O(\varepsilon^2)$, since v has degree at most $k-1$. Moreover, $\operatorname{Re}\langle v, a\rangle = \operatorname{Re}\langle v, g^*\rangle = \operatorname{Re}\langle\varphi, |h|^2\rangle = 0$. With $Y = X_\varepsilon - a$ we therefore have $\operatorname{Re}\langle Y, a\rangle = O(\varepsilon^2)$ and $\|Y\|_{H^2}^2 = q^2\|v\|_{H^2}^2\varepsilon^2 + O(\varepsilon^3)$, and expanding $\|a+Y\|_{H^2} = (R^2 + 2\operatorname{Re}\langle Y, a\rangle + \|Y\|_{H^2}^2)^{1/2}$ we get

$$R\|X_\varepsilon\|_{H^2} = \operatorname{Re}\langle X_\varepsilon, a\rangle + \frac{q^2}{2}\|v\|_{H^2}^2\varepsilon^2 + O(\varepsilon^3).$$

In other words, the norm and its supporting functional at a agree to first order, and their difference begins with half the squared displacement. Moreover, $n_\varepsilon = 1 + \varepsilon^2\|v\|_{H^2}^2$ gives $(n_\varepsilon^{-q} - s^2)^{1/2} = R - q\|v\|_{H^2}^2\varepsilon^2/(2R) + O(\varepsilon^4)$, and $\|X_\varepsilon\|_{H^2} = R + O(\varepsilon^2)$. Inserting all of this into (8), we obtain

$$\begin{aligned}\Lambda(h_\varepsilon) = \operatorname{Re}\big(s(c_k - \mu s)\big) + \mu\big(s\operatorname{Re}[z^k]B_\varepsilon + \operatorname{Re}\langle X_\varepsilon, a\rangle\big) \\ + \frac{q(q-1)\mu}{2}\|v\|_{H^2}^2\varepsilon^2 + O(\varepsilon^3).\end{aligned}$$

The prescribed constant term and the optimized remaining coefficients now combine into one inner product. Indeed, since $g^* = a + sz^k$ with s real and g^* has degree at most k,

$$s\,\mathrm{Re}[z^k]B_\varepsilon + \mathrm{Re}\langle X_\varepsilon, a\rangle = \mathrm{Re}\langle B_\varepsilon, g^*\rangle = \mathrm{Re}\int_0^{2\pi}(1+\varepsilon\varphi)^q|h|^2\,\frac{\mathrm{d}\theta}{2\pi}$$
$$= 1 + \frac{q(q-1)}{2}\,\mathrm{Re}\langle\varphi^2, |h|^2\rangle\varepsilon^2 + O(\varepsilon^3),$$

where we used that $|g^*| = |h|$ on $\mathbb{T}$, that $\|h\|_{H^2} = 1$ and that $\mathrm{Re}\langle\varphi, |h|^2\rangle = 0$. At $\varepsilon = 0$ this gives $\Lambda(h) = \mathrm{Re}(s(c_k - \mu s)) + \mu$, and so

$$\Lambda(h_\varepsilon) = \Lambda(h) + \frac{q(q-1)\mu}{2}\Big(\|v\|_{H^2}^2 + \mathrm{Re}\langle\varphi^2, |h|^2\rangle\Big)\varepsilon^2 + O(\varepsilon^3).$$

Finally, $\|v\|_{H^2}^2 = \langle|\varphi|^2, |h|^2\rangle$ and $|\varphi|^2 + \mathrm{Re}\,\varphi^2 = 2(\mathrm{Re}\,\varphi)^2$, which gives (7). □

We are now ready to prove our first main result.

Proof of Theorem 1. For $t = 1$ the only admissible function is $f = 1$. Let $0 \le t < 1$, and suppose that an extremal $f = gh^q$ has two zeros in $\mathbb{D}$. Then g is not constant, and the direction $\varphi = \omega z/W$ constructed above satisfies the hypotheses of Lemma 2. Since $\varphi \neq 0$, this is a contradiction. □

4. The terminal interval

Throughout this section $k \ge 1$ is fixed, and constants depend only on p. Put $m = \lceil 2/p\rceil$ and $r = mp$, so that $m \ge 3$ and $2 \le r < 3$. Every admissible f in (1) with $t > 0$ factors as

$$f = F_1 \cdots F_m, \qquad \|F_i\|_{H^r} \le 1, \qquad a_i = F_i(0) > 0, \qquad a_1 \cdots a_m = t,$$

by placing the inner factor of f in F_1, taking equal mth roots of its outer factor, and adjusting phases. Since $a_i \le \|F_i\|_{H^r} \le 1$, we have $t \le a_i \le 1$. The exponent $r \ge 2$ gives a Hilbert-space bound for products of two factors and a quadratic convexity deficit, while $r < 3$ ensures that products of three or more factors lie in H^κ with $\kappa < 1$.

For $F(0) = a$, the model maximizer is the binomial $Q = a(1+\beta z^k)^{2/r}$ with $\beta = \sqrt{a^{-r} - 1}$, which has norm one and coefficient $2a\beta/r$ of z^k. The next lemma is a stability estimate. It states that losing little in the coefficient of z^k forces a function to be close to this binomial in both H^2 and H^r. The binomial works as a comparison function because its norm derivative, the boundary function $|Q|^{r-2}\overline{Q}$, has only one negative Fourier frequency, namely $-k$. Paired with an analytic deviation vanishing at 0, it therefore detects exactly the coefficient being maximized.

Lemma 3. *Fix $2 \le r < 3$. There are constants $a_r < 1$ and $c_r > 0$ with the following property. Suppose that $F \in H^r$ with $\|F\|_{H^r} \le 1$ and $a = F(0) \in [a_r, 1)$, put $\beta = \sqrt{a^{-r} - 1}$, choose $|\eta| = 1$ with $\overline{\eta}[z^k]F \ge 0$, and write*

$$F = Q + W, \qquad Q(z) = a(1 + \eta\beta z^k)^{2/r}.$$

Then

$$|[z^k]F| \le \frac{2a\beta}{r} - c_r\,\frac{e}{\beta}, \qquad e = \|W\|_{H^2}^2 + \|W\|_{H^r}^r. \tag{9}$$

In particular, $e \le \beta^2/c_r$.

Proof. We begin with a pointwise inequality. For complex v, w with $1/2 \le |v| \le 2$ we have

$$|v+w|^r \ge |v|^r + r\,\mathrm{Re}\big(|v|^{r-2}\overline{v}w\big) + c\big(|w|^2 + |w|^r\big)$$

with $c > 0$ depending only on r. Indeed, the difference of the two sides without the last term is positive for $w \neq 0$ by the strict convexity of $|\cdot|^r$, it is at least a positive multiple of $|w|^2$ for small w because the real Hessian of $|\cdot|^r$ is positive definite away from 0, and it is asymptotic to $|w|^r$ for large w. All of this is uniform on the annulus.

Choose a_r so that $1/2 \le |Q| \le 2$ on $\mathbb{T}$ when $a \ge a_r$. Since $|Q|^r = a^r|1+\eta\beta z^k|^2$ on $\mathbb{T}$, we have $\|Q\|_{H^r}^r = 1$. On $\mathbb{T}$, we also have

$$|Q|^{r-2}\overline{Q} = a^{r-1}(1+\eta\beta z^k)^{1-2/r}(1+\overline{\eta}\beta\,\overline{z}^k),$$

whose only negative Fourier coefficient is $a^{r-1}\overline{\eta}\beta$, at frequency $-k$. Since $W(0) = 0$, integrating against W picks out the coefficient of z^k in W, and $\overline{\eta}[z^k]W = |[z^k]F| - 2a\beta/r$. Integrating the pointwise inequality with $v = Q$ and $w = W$ therefore gives

$$1 \ge \|F\|_{H^r}^r \ge 1 + ra^{r-1}\beta\Big(|[z^k]F| - \frac{2a\beta}{r}\Big) + ce,$$

which is (9) with $c_r = c/r$, since $a \le 1$. The last assertion follows because the left-hand side of (9) is nonnegative and $2a/r \le 1$. □

Armed with Lemma 3, we can now prove our second main result.

Proof of Theorem 2. For $t = 1$ the only admissible function is 1. Let $0 < t < 1$ with $D = 1 - t$ small, and factor an admissible f as above. All the a_i lie in $[t, 1]$, so for D small enough Lemma 3 applies to every F_i with $a_i < 1$. If $a_i = 1$, then $F_i \equiv 1$, and we put $W_i = 0$. Write $F_i = Q_i + W_i$, e_i and $\beta_i = \sqrt{a_i^{-r} - 1}$ as in the lemma, and put $E = \sum_i e_i$. Since $a_i \ge t$, we have $\beta_i^2 \le t^{-r} - 1 \le C_1 D$, and hence $E \le C_1 D/c_r$. Here and below $C_1, C_2, \ldots$ and $c_1, c_2, \ldots$ denote positive constants depending only on p.

Step 1: terms with one nonconstant factor. Expanding each $F_i = a_i + (F_i - a_i)$, we see that the terms of $[z^k]f$ that use exactly one nonconstant part are $t\sum_i [z^k]F_i/a_i$. By (9) and $a_i \le 1$,

$$t\sum_i \frac{|[z^k]F_i|}{a_i} \le \frac{2t}{r}\sum_i \sqrt{a_i^{-r} - 1} - \frac{c_1 E}{\sqrt{D}}.$$

The function $x \mapsto \sqrt{e^{rx} - 1}$ is concave for $e^{rx} \le 2$, and $x_i = -\log a_i$ satisfies $\sum_i x_i = -\log t$. If $t^{-r} \le 2$, Jensen's inequality gives $\sum_i \sqrt{a_i^{-r} - 1} \le$

$m\sqrt{t^{-r/m}-1}$, with equality only if all the a_i are equal. Since $r/m = p$ and $m/r = 1/p$, we find that

$$t\sum_i \frac{|[z^k]F_i|}{a_i} \le \frac{2t}{p}\sqrt{t^{-p}-1} - \frac{c_1 E}{\sqrt{D}}.$$

The first term is $(2\beta/p)(1+\beta^2)^{-1/p}$, which is the value of the binomial $f_\beta = (1+\beta z^k)^{2/p}(1+\beta^2)^{-1/p}$, and f_β is admissible.

Step 2: terms with several nonconstant factors. Each $Q_i - a_i$ is a power series in z^k without constant term, so a product of at least two nonconstant parts can contribute to the coefficient of z^k only if all of them are W_i's. For two factors, the Cauchy–Schwarz inequality gives $|[z^k]W_iW_l| \le \|W_i\|_{H^2}\|W_l\|_{H^2} \le E$. For a set J of $j \ge 3$ factors, put $\kappa = r/j < 1$. By Hölder's inequality the product lies in H^κ with norm at most $\prod_{i\in J}\|W_i\|_{H^r}$, and the coefficients of H^κ functions satisfy $|[z^k]V| \le C_\kappa k^{1/\kappa-1}\|V\|_{H^\kappa}$ [5, Theorem 6.4]. Since $\|W_i\|_{H^r} \le e_i^{1/r}$, $j/r > 1$ and $E \le C_1 D/c_r$, we get

$$\Big|[z^k]\prod_{i\in J} W_i\Big| \le C_2 k^{j/r-1}E^{j/r} \le C_3 k^{j/r-1}D^{j/r-1}E.$$

The constant factors a_i multiplying these products are at most 1.

Step 3: comparison. Each term from Step 2 is $\sqrt{D}\,(kD)^{j/r-1}E/\sqrt{D}$ with $0 < j/r - 1 \le 1/p - 1$, up to a constant, and the two-factor terms are at most a constant times $\sqrt{D}\,E/\sqrt{D}$. After distinguishing between $kD \le 1$ and $kD \ge 1$, we see that all of them together are at most a constant times $(\sqrt{D} + k^{1/p-1}D^{1/p-1/2})E/\sqrt{D}$. Adding Step 1, we obtain

$$\mathrm{Re}[z^k]f \le \frac{2t}{p}\sqrt{t^{-p}-1} - \frac{E}{\sqrt{D}}\Big(c_1 - C_4(\sqrt{D} + k^{1/p-1}D^{1/p-1/2})\Big).$$

The quantity $k^{1/p-1}D^{1/p-1/2}$ is the ratio between the concentration value $D^{1/p}k^{1/p-1}$ and the binomial value $\sqrt{D}$ from the introduction, and it is small exactly when $D \ll k^{-\alpha}$. Taking c small, the bracket is positive whenever $D \le ck^{-\alpha}$, for every $k \ge 1$. Then $\mathrm{Re}[z^k]f$ is at most the value of f_β, which proves the value in (a). The case $k = 1$ gives $\Phi_1(p,t)$ the same value.

For uniqueness, suppose that f attains this value. Then $E = 0$, since the bracket is positive, so $F_i = Q_i$ for every i. Equality in Jensen's inequality forces all the a_i to equal $t^{1/m}$, and equality in $\mathrm{Re}[z^k]F_i \le |[z^k]F_i|$ forces every $\eta_i = 1$. Hence $f = \prod_i t^{1/m}(1+\beta z^k)^{2/r} = t(1+\beta z^k)^{2/p} = f_\beta$ with $\beta = \sqrt{t^{-p}-1}$, which completes the proof of (a).

Step 4: concentration. For (b), let $n = k - 1 \ge 2$ and $\rho = 1 - 1/n$. The function $K(z) = (1-\rho^2)^{1/p}(1-\rho z)^{-2/p}$ has norm 1 in H^p, since $|K|^p$ is the Poisson kernel at ρ, and its coefficient of z^n is $(1-\rho^2)^{1/p}\rho^n\Gamma(n + 2/p)/(\Gamma(2/p)\,n!) \ge c_2 n^{1/p-1}$. Since $|x+y|^p \le |x|^p + |y|^p$, the function $t + (1-t^p)^{1/p}zK$ is admissible, and since $1 - t^p \ge p(1-t)$, we get

$$\Phi_k(p,t) \ge c_3(1-t)^{1/p}k^{1/p-1}.$$

On the other hand, by (a) with $k = 1$, we have $\Phi_1(p,t) \le C_5\sqrt{1-t}$ for t close to 1. With $1 - t = Ck^{-\alpha}$, the ratio of the two bounds is $(c_3/C_5)C^{1/p-1/2}k^{(1/p-1)-\alpha(1/p-1/2)} = (c_3/C_5)C^{1/p-1/2}$, since the exponent of k vanishes. Taking C large makes it exceed 1, and t is close to 1 when k is large. This proves (b). □

5. Proof of Corollary 1

Proof of Corollary 1. For $t = 0$, every function f in (1) vanishes at 0, so by Theorem 1 the extremals have exactly one zero. For $t > 0$, it suffices by Theorem 1 to show that the extremals have a zero when t is small. Since $|a+b|^p \le |a|^p + |b|^p$ for $0 < p < 1$, the function $t + (1-t^p)^{1/p}z^k$ is admissible in (1), so $\Phi_k(p,t) \ge (1-t^p)^{1/p} \to 1$ as $t \to 0$. We argue by contradiction. Suppose that there are $0 < t_n < 1$ with $t_n \to 0$ and extremals f_n for $\Phi_k(p,t_n)$ without zeros in $\mathbb{D}$. Since $\|f_n\|_{H^p} = 1$, the f_n are uniformly bounded on compact subsets of $\mathbb{D}$, so a subsequence converges locally uniformly to some f. By Hurwitz's theorem f is either zero-free or identically zero, and since $f(0) = \lim f_n(0) = 0$, we have $f = 0$. But then the coefficients of z^k of the f_n tend to 0 along the subsequence, which contradicts $\operatorname{Re} f_n^{(k)}(0)/k! = \Phi_k(p,t_n) \ge (1-t_n^p)^{1/p}$. □

6. Concluding remarks and further work

6.1. **Real versus complex directions.** If one averages (8) over complex $\varepsilon = re^{i\theta}$, as in the proof of [3, Lemma 3], the terms involving $\operatorname{Re}\langle\varphi^2, |h|^2\rangle$ disappear, and the resulting inequality does not seem to exclude two zeros. The restriction to a real direction in Lemma 2 is therefore essential to our proof of Theorem 1.

6.2. **The leading constant.** The question left unaddressed in this paper is the precise asymptotic behaviour of the terminal threshold $T_k(p)$. The constants in Theorem 2 are not sharp. Write A_p for the limit of $n^{1-1/p}C(n,p)$, assuming that it exists, and $q = 2/p-1$. Optimizing the split of the remaining norm between the binomial and a concentrated profile, as in the heuristic of the introduction, leads to the following.

Conjecture 1. *For $0 < p < 1$,*

$$k^{\alpha}\bigl(1 - T_k(p)\bigr) \longrightarrow \frac{1}{p}\Bigl(\frac{q^q}{A_p(q^2-1)^{(q-1)/2}}\Bigr)^{2/q} \qquad (k \to \infty).$$

Acknowledgements. AI assistance was used in developing and checking the arguments and in preparing the exposition.

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